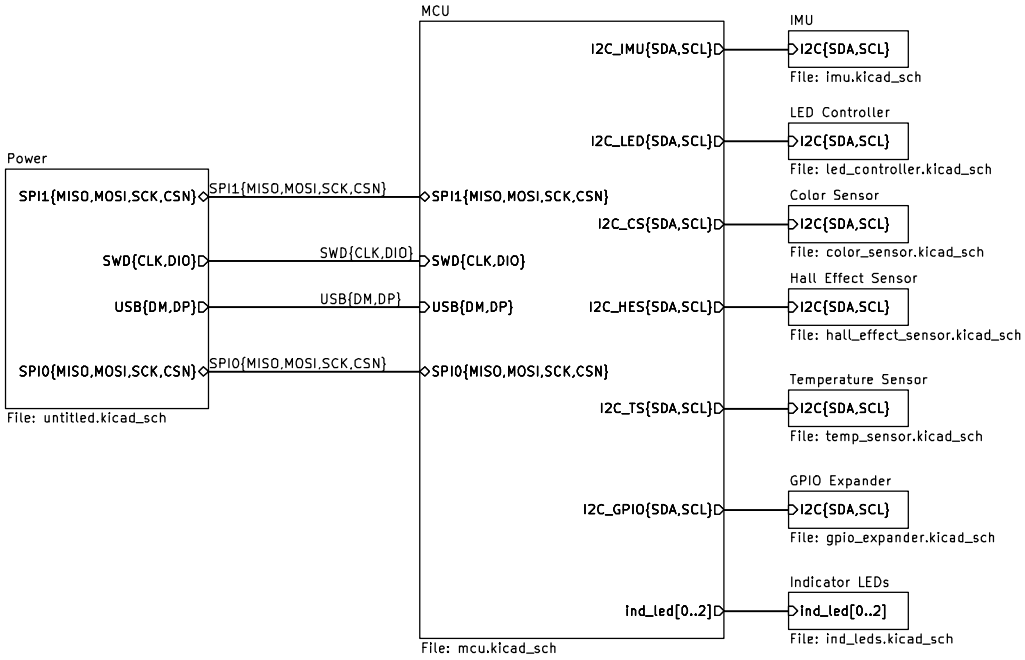
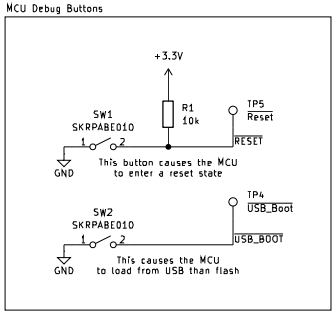
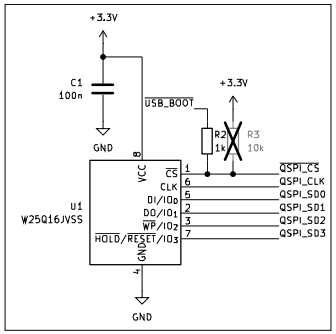

Power
SPI1{MISO,MOSI,SCK,CSN}
SWD{CLK,DIO}
USB{DM,DP}
SPI0{MISO,MOSI,SCK,CSN}
File: untitled.kicad_sch
SPI1{MISO,MOSI,SCK,CSN}
SWD{CLK,DIO}
USB{DM,DP}
SPI0{MISO,MOSI,SCK,CSN}
MCU
SPI1{MISO,MOSI,SCK,CSN}
SWD{CLK,DIO}
USB{DM,DP}
SPI0{MISO,MOSI,SCK,CSN}
I2C_IMU{SDA,SCL}
I2C_LED{SDA,SCL}
I2C_CS{SDA,SCL}
I2C_HES{SDA,SCL}
I2C_TS{SDA,SCL}
I2C_GPIO{SDA,SCL}
Ind_led[0..2]
File: mcu.kicad_sch
IMU
I2C{SDA,SCL}
File: imu.kicad_sch
LED Controller
I2C{SDA,SCL}
File: led_controller.kicad_sch
Color Sensor
I2C{SDA,SCL}
File: color_sensor.kicad_sch
Hall Effect Sensor
I2C{SDA,SCL}
File: hall_effect_sensor.kicad_sch
Temperature Sensor
I2C{SDA,SCL}
File: temp_sensor.kicad_sch
GPIO Expander
I2C{SDA,SCL}
File: gpio_expander.kicad_sch
Indicator LEDs
Ind_led[0..2]
File: ind_leds.kicad_sch



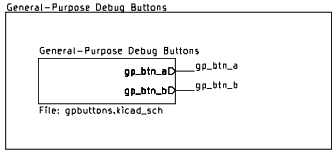
MCU Debug Buttons



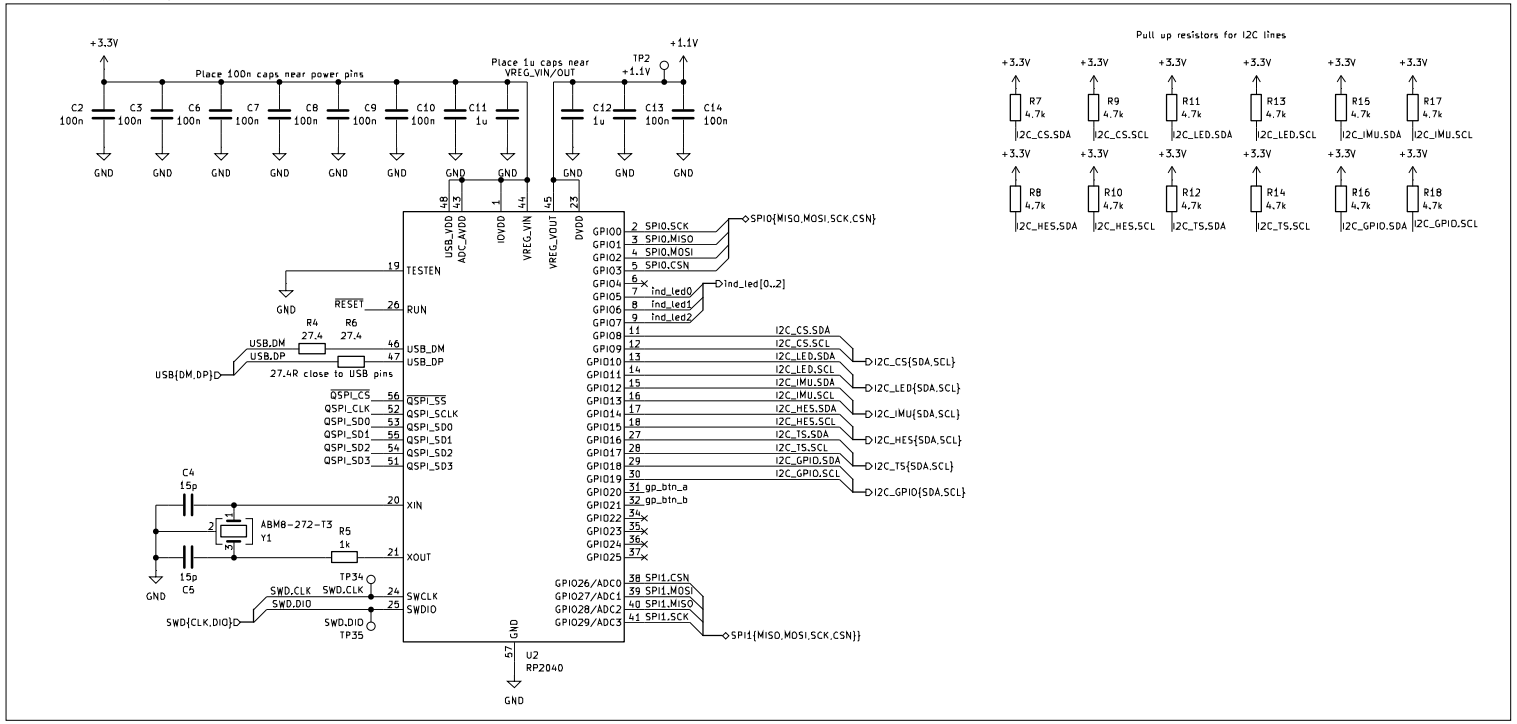
2 MB Flash



General-Purpose Debug Buttons

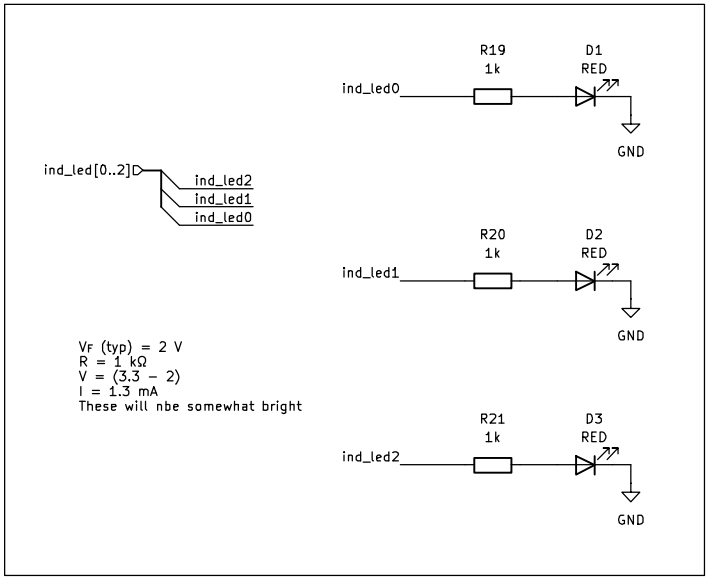


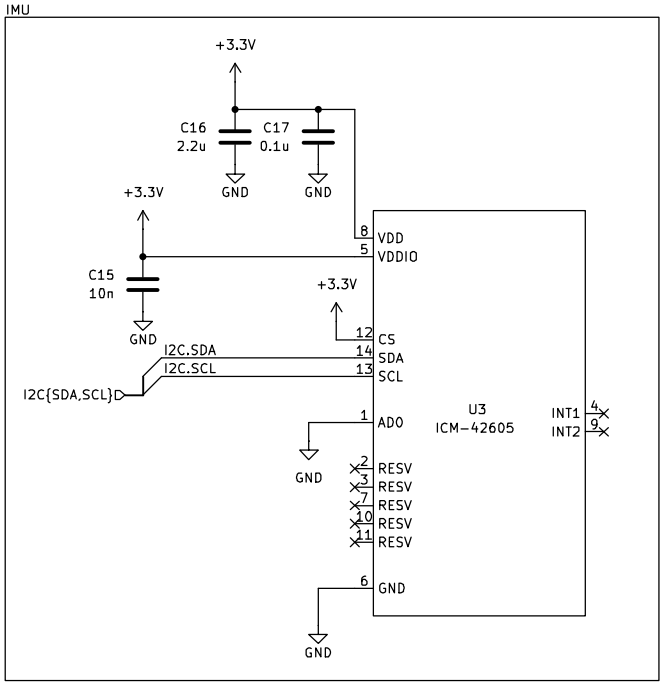
RP2040 MCU + Support Circuitry



Pull up resistors for I2C lines

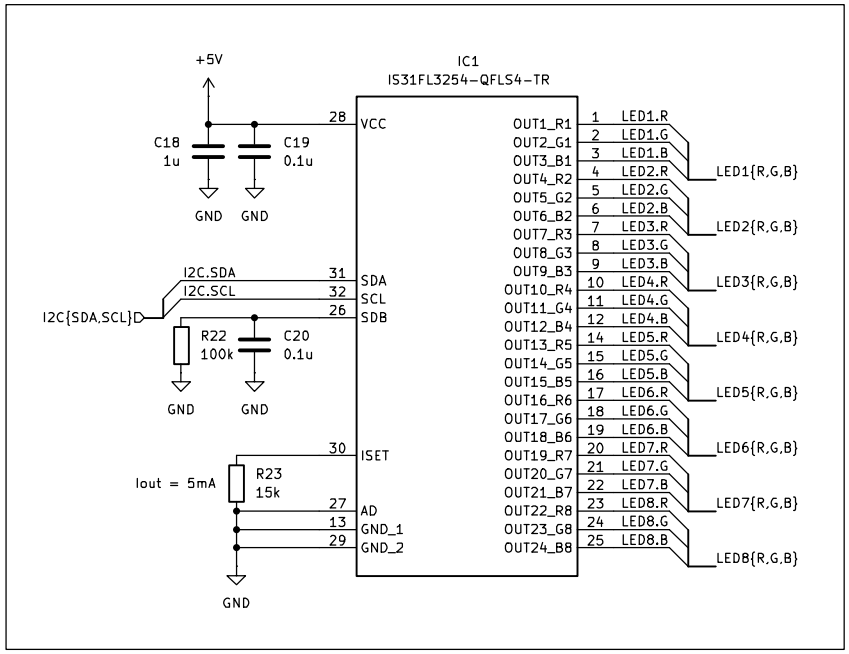
Indicator LEDs





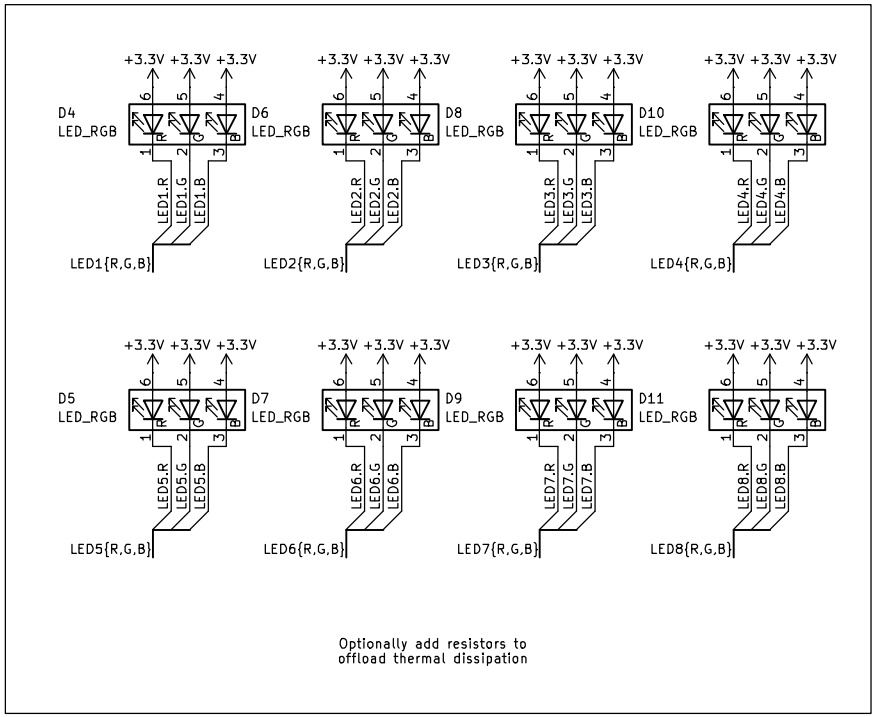
Device	ICM-42605 (U8)
Function	Interial Measurement Unit
I2C Address (7-bit)	1101000 (0x68)
Address Config	AD0 on GND
Voltage Range	1.7-3.6V

LED Driver

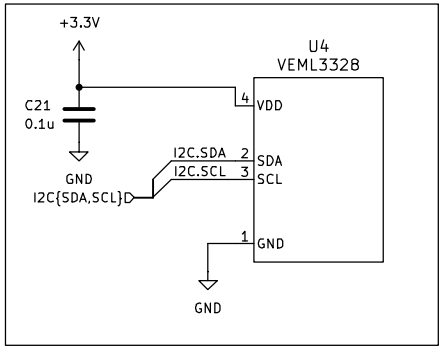


Device	IS31FL3254-QFLS4-TR (IC1)
Function	LED Controller
I2C Address (7-bit)	0110000 (0x30)
Address Config	AD on GND
Voltage Range	2.7-5.5V
Pull-Ups	4.7k on I2C
Led 0 Pin	RP2040 GPIO 5 (Pin 7)
Led 1 Pin	RP2040 GPIO 6 (Pin 8)
Led 2 Pin	RP2040 GPIO 7 (Pin 9)

RGB LEDs



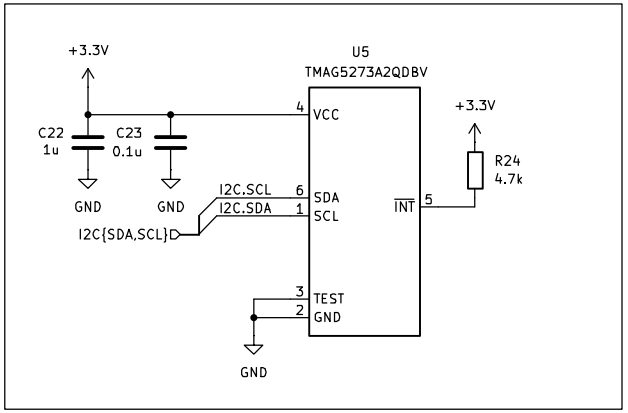
Color Sensor



Device	VEML3328 (U4)
Function	Color Sensor
I2C Address (7-bit)	0101000 (0x28)
Address Config	NONE
Voltage Range	2.6-3.6V

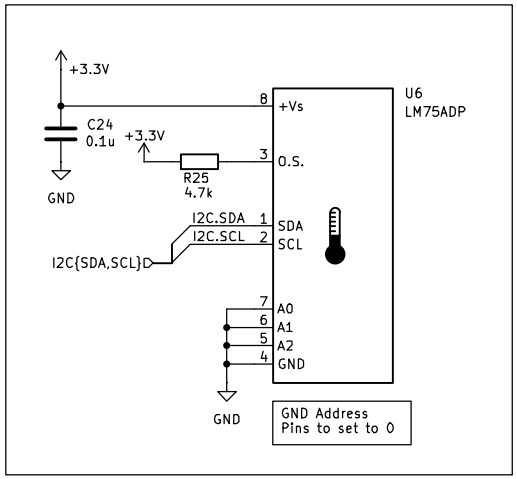
Device	TMAG5273A2 (U5)
Function	Hall Effect & Temperature Sensor
I2C Address (7-bit)	0110101 (0x35)
Address Config	Register 0x0C
Voltage Range	1.7–3.6V
Pull-Ups	4.7k on INT and I2C

Hall Effect Sensor



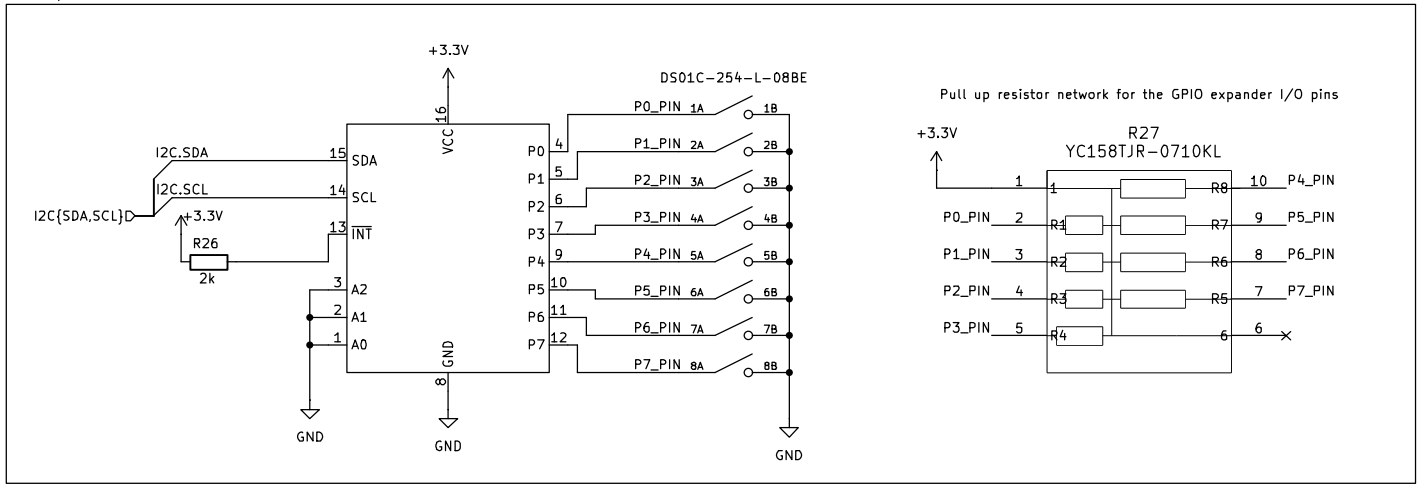
Datasheet advises to ground TEST

Temperature Sensor



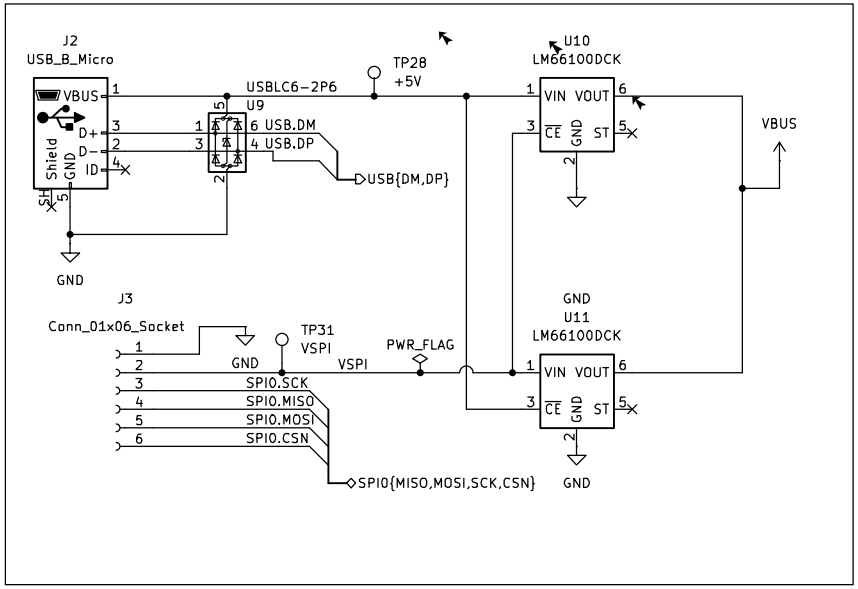
Device	LM75ADP (U6)
Function	Temperature Sensor
I2C Address (7-bit)	1001000 (0x48)
Address Config	A0-A2 on GND
Voltage Range	2.8-5.5V
Pull-Ups	4.7k on OS and I2C

GPIO Expander

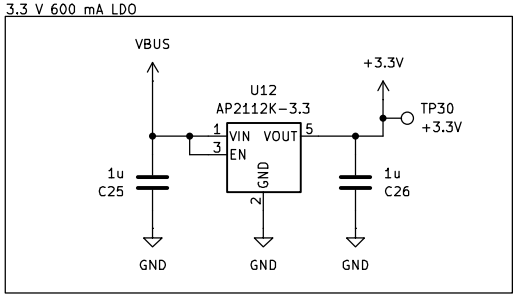


Device	TCA9534PWR (U7)
Function	GPIO Expander
I2C Address (7-bit)	0100000 (0x20)
Address Config	A0-A2 on GND
Voltage Range	1.65-5.5V
Pull-Ups	4.7k on I2C, 2K on INT to VCC

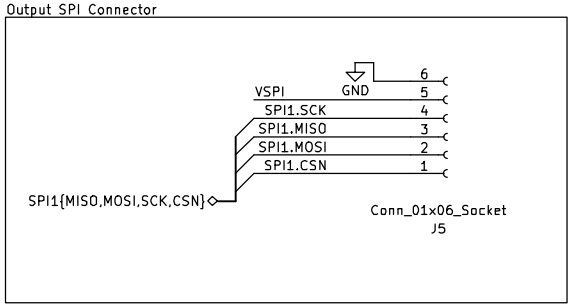
USB-C and SPI Input Power MUXing



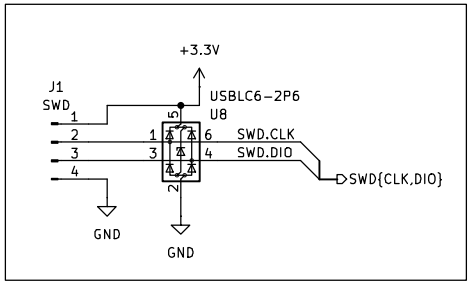
3.3 V 600 mA LDO



Output SPI Connector



SWD Debug Connector



Debug and Option Buttons

